Nominal Budgets and Realized Resources in Closed-API Large-Language-Model Inference: A Performance Evaluation Protocol

Soroush Vahidi[a,1,*]

[a] *Ying Wu College of Computing, New Jersey Institute of Technology, Newark, New Jersey, USA*

Abstract

Closed-API large language model (LLM) evaluations often match systems by nominal inference budget, but equal nominal budgets need not imply equal realized resources or fair attribution between answer discovery and final-answer selection. We present a unified protocol for closed-API, budgeted inference that records componentwise resources, evaluates identical-pool selector controls, labels provider transfer for failure-mined rules, preserves protocol-blocked outcomes, and discloses incomplete telemetry. We exercise it on 15 completed provider-by-dataset cells with 3,394 paired examples; Fireworks×GPQA-Diamond is one protocol-blocked cell. On the shared four-generator pool, identical-pool plurality (Pooled-4) exceeds the failure-trace gated selector (FTA), 66.53% versus 65.00% (McNemar p=0.00027). FTA's dominant gate transfers poorly in Azure mathematics cells (2/18 and 7/31 rescue/regression patterns). Resource reconstruction changes efficiency interpretation: Frontier successful completions rise from 2.78 to 5.38 of nominal budget B=6, while token and dollar fields remain lower bounds. Nominal budget alone is therefore insufficient for defensible closed-API inference-performance conclusions.

Keywords: budgeted inference, performance measurement, resource accounting, closed-API inference, large language models

*Corresponding author

Email address: sv96@njit.edu (Soroush Vahidi)

[1]Sole author. ORCID: 0000-0003-1934-6282.

1. Introduction

Budgeted test-time inference is often summarized by one performance number: final accuracy under a shared nominal budget of samples, branches, or reasoning actions. For performance evaluation, that summary is incomplete because the nominal budget is an experimental entitlement, not an observed resource measurement. Accuracy under the same entitlement can therefore conflate several distinct questions:

- Did any correct answer enter the candidate pool?
- Did the commit rule select a correct answer when one was present?
- Did compared methods share an identical pool, or did one simply receive more diverse candidates?
- Does a failure-derived selection rule that helps under one commercial API transfer to another?
- Does a matched nominal budget B imply comparable calls, tokens, latency, or cost?

The distinction matters because equal nominal budgets need not induce equal realized resources. A method may spend the same logical budget while producing different numbers of successful completions, retries, generated tokens, observed latency, or estimated cost. When those quantities are collapsed into final accuracy, a selector improvement can be confused with better candidate discovery, and a budget-matched comparison can be mistaken for a resource-matched one.

Closed commercial APIs make the measurement problem harder. Providers expose different operational behavior, telemetry fields may be incomplete, and hosted model identifiers do not guarantee immutable weight-level versions. A post-generation rule can also look effective when compared only to a single generator, then lose to plurality on the *same* extracted answers. Failed or unscorable provider–dataset cells create another reporting choice: they can be imputed, dropped, or preserved as protocol outcomes.

Prior work contains several relevant ingredients individually: coverage and pass@k analyses, fixed-pool diagnostics, and self-consistency or other

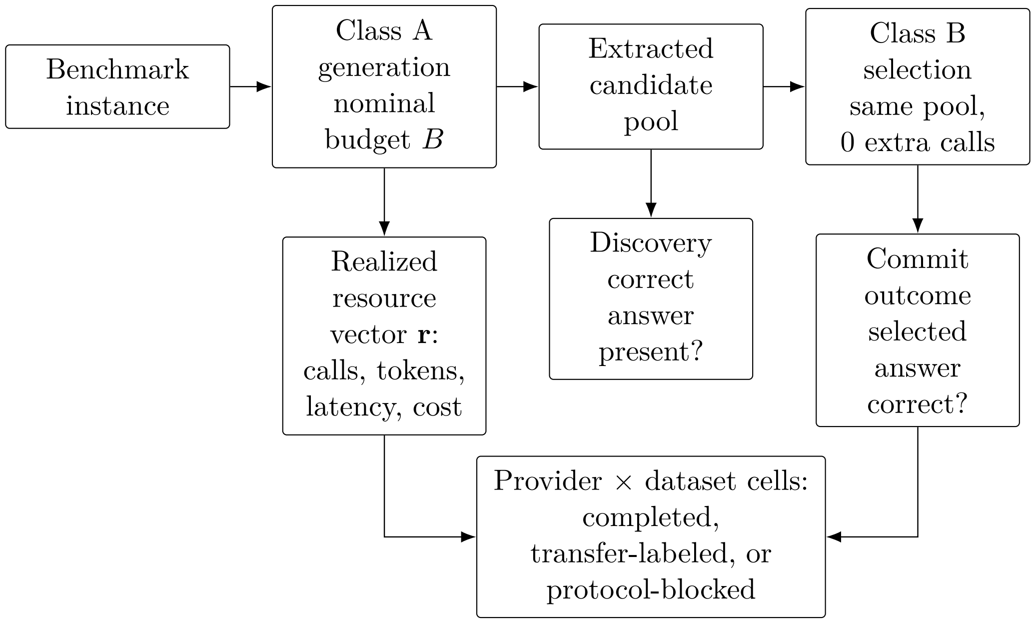

Figure 1: Protocol schematic. Nominal budget B defines the logical action contract; realized resources are measured separately. Selection claims are attributed after candidate availability is fixed, and provider×dataset cells may be completed, transfer-labeled, or protocol-blocked.

consensus aggregators [1, 2, 3, 4]. Router and test-time-compute work also motivates resource reporting under budgeted inference [5]. The measurement gap is not the absence of any one ingredient. It is the lack of one executable and auditable closed-API protocol that forces these separations to be reported together for the same provider–dataset matrix.

Contributions. This article contributes a controlled performance-evaluation protocol for budgeted closed-API inference (Figure 1). The contribution is the joint accounting contract: the same evaluation records nominal entitlements, realized resources, discovery, selector attribution, provider-transfer labels, provenance tiers, protocol-blocked cells, and telemetry completeness, so each performance conclusion has an explicit measurement layer. Specifically:

- **Protocol formulation.** We define nominal budget, candidate-pool discovery, conditional selection accuracy, consensus regret, realized resource vectors, and protocol-blocked outcomes for closed-API inference.

- **Selector attribution and transfer analysis.** We compare nontrivial commit rules to an identical-pool plurality control and label failure-mined rules by development and transfer environment.
- **Realized-resource accounting.** We separate successful completions, retries, tokens, latency, and estimated dollars, while marking incomplete telemetry and reconstruction status.
- **Released evidence boundary.** We report the blocked cell rather than imputing it and provide aggregate-record verification artifacts for the reported tables and numerical invariants.

The empirical study exercises the protocol on four providers and four reasoning workloads at one nominal budget, with 15 completed cells and one protocol-blocked cell. It demonstrates that identical-pool plurality can change selector conclusions, that a failure-mined gate can transfer poorly across providers, and that reconstructed successful completions can change efficiency interpretation even when accuracy and paired tests are unchanged.

The study does not claim a general algorithm ranking, a complete cost-efficiency or Pareto analysis, official reproductions of L1/S1/TALE, or guaranteed regeneration of identical proprietary API outputs. Frontier, FTA, Pooled-4, and the adapters are measurement objects used to exercise the protocol, not the central contribution.

Roadmap. Section 2 positions the work relative to performance measurement, test-time compute, routing, consensus, and verifier literature. Sections 3–5 define the evaluation objects and closed-API workload. Sections 6–9 report the matrix, identical-pool selection tests, provider transfer, and resource accounting. Section 10 reports offline counterfactuals, and Sections 11–13 summarize scope and implications for performance-evaluation practice.

Measurement objects. The named procedures are measurement objects for exercising the protocol (Section 4). Class A single-generator policies under B=6 include a reserved-attempt full-expansion diagnostic (Frontier) and closed-API adapters inspired by L1, S1, and TALE [6, 7, 8]. Class B post-generation selectors over the shared pool include Pooled-4 and FTA. Offline oracles are upper bounds only. A single-cell Azure×GSM8K call/sample-budget control compares Frontier to same-model self-consistency with N=6 (SC-N=6; Section 6.3).

2. Related Work

Performance measurement and cloud benchmarking. Performance-evaluation practice emphasizes construct validity, measurement semantics, and protocol transparency: rankings depend on what is counted, at what layer, and under what workload definition [9, 10, 11]. Standardized inference benchmarking makes the same point at systems scale: MLPerf Inference specifies workloads, metrics, and reproducibility rules so architecture-neutral comparisons remain meaningful [12]. Cloud benchmarking methodology similarly calls for explicit measurement and reporting practice, while recent PEVA work on microservices and serverless ML treats latency, scaling, and cost as part of the performance object [13, 14, 15]. We inherit that measurement view but apply it to closed-API LLM inference, where the nominal budget is a logical entitlement and not the realized resource vector.

Benchmark validity and evaluation infrastructure. LLM evaluation infrastructure such as HELM emphasizes scenario coverage, multi-metric reporting, and released evaluation artifacts; its evaluation object is broad model behavior under documented scenarios rather than budget attribution [16]. LiveBench addresses benchmark drift and contamination with frequently updated, objectively scored tasks [17]. GSM-Symbolic shows that GSM8K-style math scores can be sensitive to symbolic perturbations of the same underlying problem templates [18]. These works motivate our workload-validity caveats and our choice to treat absolute accuracies as protocol snapshots. They are not direct baselines for the closed-API accounting protocol.

Test-time compute and adaptive inference. Test-time-compute work studies how inference budget affects accuracy, often by controlling generation length, sample count, or input-adaptive allocation. Surveys distinguish fixed-budget controllability from adaptive allocation, and recent methods formalize or learn budget policies under finite compute constraints [19, 5, 20, 21]. Length-control and budget-forcing systems such as L1, s1, and TALE intervene directly on model inference trajectories through training, local token-stream control, or token-budget prompting [6, 7, 8]. Our evaluated L1-/S1-/TALE-style rows are closed-API adapters used as Class A measurement objects under the same nominal-B contract, not official reproductions of those systems (Table 1).

Self-consistency, consensus, and fixed-pool selection. Pass@k analyses measure whether at least one generated sample is correct, separating candidate availability from final selection [1, 2]. Same-model self-consistency selects a majority answer from multiple samples of one model, and Universal Self-Consistency replaces extractive majority with model-mediated comparison for freer-form outputs [4, 22]. Adaptive self-consistency methods reduce sample cost by stopping or allocating samples based on agreement, difficulty, or reliability signals [23, 24, 25, 26]. We inherit the discovery/selection distinction from this line of work. Pooled-4 differs in evaluation object: it is a heterogeneous four-generator plurality control over the same extracted answers used by FTA, while the Azure GSM8K SC-N=6 experiment is a single-cell supporting control rather than a full SC benchmark.

Best-of-N, verifier selection, routing, and cascades. Best-of-N and verifier-based methods use additional signals to choose among candidates; process supervision and self-certainty are examples of such selection layers [27, 28]. Fixed-pool oracle-gap diagnostics decompose recoverable mass, signal coverage, conditional selection quality, and harm, making the candidate pool itself the evaluation object [3]. Routing and cascade systems such as FrugalGPT, Hybrid LLM, and RouteLLM choose among models to trade cost against quality [29, 30, 31]. This study does not introduce a new consensus algorithm, verifier, or learned router. It integrates identical-pool attribution, provider-transfer labels, and realized-resource accounting into one closed-API evaluation protocol.

Closed-API resource and telemetry accounting. Open serving-stack systems expose resource layers unavailable in black-box APIs. Orca studies iteration-level scheduling and selective batching for Transformer serving, while vLLM/PagedAttention optimizes KV-cache memory management and throughput in instrumented serving systems [32, 33]. Closed commercial services instead expose partial telemetry: successful calls, retries, token fields, latency, and billable cost may not align perfectly. Hosted LLM services can also change over time, making exact longitudinal regeneration difficult [34]. Our protocol therefore reports reconstruction status, lower-bound token/USD fields, provider/model identifiers, and aggregate-record verification boundaries.

Search, stopping, and metareasoning. Tree-of-Thoughts expands intermediate states under search heuristics, and classical metareasoning frames stopping as a value-of-computation decision [35, 36]. The evaluated Frontier configuration is not an adaptive early-stopping algorithm: its retained early-commit gate is disabled by sentinel thresholds, and the policy is used here as a Class A accounting diagnostic (Section 4.2). Historical search variants outside the evaluated protocol are documented only in the supplementary material.

3. Evaluation Framework

Setting. The protocol deliverable is a set of accounting and attribution rules for budgeted closed-API inference. An instance x_i has gold answer g_i, used offline only for evaluation. A single-generator method may issue at most B logical reasoning actions (provider calls) while producing a committed answer. We call B the *nominal budget*: the logical entitlement specified by the experiment, not an observed resource total. A candidate pool $C_i = \{y_{i,1}, \ldots, y_{i,m}\}$ is a multiset of extracted final answers under a shared protocol. A selector f maps C_i and optional gold-free metadata to a prediction $\hat{y}_i$.

Discovery and selection. Following coverage / pass@k analyses [1, 2] and fixed-pool recoverable-mass diagnostics [3], define the discovery indicator

$$D_i \;=\; \mathbf{1}\{\exists\, y \in C_i :\; y \equiv g_i\}, \tag{1}$$

i.e., whether a correct candidate is present in the evaluated pool. With n scored instances and $N_D = \sum_i D_i$, define discovery rate (DR), unrecoverable failure rate (UFR), conditional selection accuracy (CSA_f), and selection failure rate (SFR_f):

$$\begin{aligned} \mathrm{DR} &= \tfrac{1}{n}\sum_i D_i, & \mathrm{UFR} &= 1 - \mathrm{DR}, \\ \mathrm{CSA}_f &= \frac{\sum_i \mathbf{1}\{\hat{y}_i \equiv g_i\}\, D_i}{N_D} \quad (N_D > 0), & \mathrm{SFR}_f &= 1 - \mathrm{CSA}_f. \end{aligned} \tag{2}$$

Blocked or unscorable instances are excluded from n rather than counted as $D_i = 0$ (Section 5). For pool-only selectors under exact-match scoring,

$$\mathrm{Acc}_f \;=\; \mathrm{DR} \times \mathrm{CSA}_f, \tag{3}$$

because such a selector can be correct only when $D_i = 1$.

Identical-pool consensus. Let $\mathrm{Cons}(C_i)$ denote plurality over C_i, with ties broken by a fixed documented rule. An *identical-pool control* holds C_i fixed and changes only the commit rule. Consensus regret $\mathrm{CR}(f) = \mathrm{Acc}(\mathrm{Cons}) - \mathrm{Acc}(f)$ is positive when consensus beats f on the same pools. We report CR whenever a nontrivial selector is compared [4, 22]. Offline oracles over C_i are Class C upper bounds only [3].

Realized compute vector. Nominal budget B is an experimental contract, not a realized-resource metric. It counts logical Class A reasoning actions, not dollars or FLOPs. A *successful completion* is a provider call that returns a usable model output on the reported path; a *retry* is an internal repeated HTTP attempt after a transient or malformed response. Under the nominal budget we record a componentwise realized resource vector

$$\mathbf{r} \;=\; (\text{calls},\ \text{generated tokens},\ \text{latency},\ \$), \tag{4}$$

distinguishing successful completions from retries, generated-token fields, observed latency, and estimated dollar cost. We do not collapse $\mathbf{r}$ to one scalar. For post-generation selectors, pool cost is the joint cost of the generators that form C, plus zero marginal selector calls. Equal B equalizes Class A opportunity, not tokens, latency, or dollars (Section 9).

3.1. Evaluation objects

Equations (1)–(4) reuse established constructs. The evaluation objects of this study are the joint reporting of DR, CSA, CR, and componentwise $\mathbf{r}$; the separation of Class A (allocation under B) from Class B (selection over a shared pool); development-environment labels for failure-mined rules; and blocked cells as *protocol-blocked outcomes*, meaning bounded-protocol failures reported separately from scored accuracy. Section 4 specifies the procedures used to instantiate those objects.

4. Measurement Instruments

The procedures below instantiate the evaluation questions of Sections 1 and 3: how nominal budgets map to realized resources; how much of a result is discovery versus commit-rule selection; and whether failure-mined commit semantics transfer across providers. The hierarchy is evaluation question → control/instrument → evidence. Names such as Frontier, FTA, Pooled-4, and the adapters identify those controls in tables.

4.1. Taxonomy

- **Class A — single-generator methods (nominal B=6 each):** Frontier; the L1-style, S1-inspired, and TALE-EP-style closed-API adapters.
- **Class B — selectors over the shared multi-generator pool:** Pooled-4 (plurality); FTA (failure-trace gated deferral). Zero *additional* model calls after pool generation; pool construction may cost up to $4B$ nominal generator actions.
- **Class C — offline oracles:** coverage upper bounds. Never treated as peer winners.

Nominal budget B is the logical entitlement of Section 3; realized $\mathbf{r}$ is tracked separately.

4.2. Frontier (Class A diagnostic)

Frontier is the Class A accounting diagnostic, not a production algorithm recommendation: a reserved-attempt full-expansion policy with fixed parameters across cells. It runs two direct-chain attempts, forms a gold-free incumbent by plurality, then uses the remaining nominal budget for challenger expansion/verification before applying a fixed gold-free override rule. On preserved validated records, reconstructed successful calls average 5.38 of nominal 6 (Section 9).

The implementation template retained an early-commit gate with sentinel thresholds $(2.0, 2.0, -1.0)$. Because top support is a vote-share in $[0, 1]$, the gate never fires; reported runs therefore do not evaluate adaptive early stopping. Full pseudocode and action semantics appear in Appendix F.

4.3. Class B selectors

Pooled-4. Plurality over $\{\text{Frontier}, \text{L1}, \text{S1}, \text{TALE}\}$ (agreement ≥ 2; ties fall back to Frontier). Same extracted answers as FTA; no additional model calls after pool generation. Its role is selector attribution, not a same-model SC replacement.

FTA (gated deferral diagnostic). FTA retains Frontier's answer or defers using Frontier trace metadata. Gates fire in precedence order; at most one per instance (Appendix G):

- **Low-Depth Guard (FIX-2).** On a single-weak-branch (or support-zero) signature, replace Frontier by majority among $\{\mathrm{L1}, \mathrm{S1}, \mathrm{TALE}\}$.
- **External-Consensus Fallback (FIX-4).** Else, if Frontier committed via direct incumbent agreement and all three externals unanimously disagree, adopt that unanimous answer.

FTA is the transfer diagnostic for failure-mined commit semantics, not a learned router. Gate rules were developed on Cohere×GSM8K failure analysis; that cell is development-adjacent (Section 8). Runtime features are gold-free (Appendix A).

4.4. Bounded answer repair

A deterministic support-based re-ranking is computed for every method but overridden by Frontier's controller-committed answer when present, while L1/S1/TALE typically score the repair layer. Offline quantification and winner-stability checks appear in Appendix B; primary tables retain the historical scoring convention.

5. Experimental Protocol

5.1. Providers, datasets, and shared protocol

Providers and exact hosted model IDs:

- Cohere: `command-r-plus-08-2024`.
- Azure OpenAI: `gpt-4.1-mini`.
- Google Vertex Gemini: `gemini-2.5-flash`.
- Fireworks: `accounts/fireworks/models/deepseek-v4-pro`.

Datasets and sizes per completed cell:

- GSM8K [37]: n=300 examples under seed 71 and nominal B=6.

- MATH-500 [38, 27]: the 500-problem MATH subset popularized in process-supervision work; we evaluate n=300 examples under the same seed/B protocol.
- GPQA-Diamond [39]: n=198 (full Diamond set under our MCQ scoring protocol).
- StrategyQA [40]: n=100. This sample is lower-powered for detecting small effects; we treat small StrategyQA deltas as descriptive, not as strong evidence.

Total paired examples across completed cells: 3,394. The evaluated protocol uses a single nominal budget B=6.

Answer scoring. Final answers use dataset-appropriate normalization. Gold labels are offline-only. Closed commercial APIs; public-benchmark contamination cannot be ruled out, so absolute accuracies are protocol snapshots. Defaults unless overridden: temperature 0.1, max output tokens 220, up to 5 API retries (base 1s, backoff ×2, max 20s, jitter 0.5s).

5.2. Matrix completeness and protocol-blocked outcome

Fifteen of sixteen provider×dataset cells completed. Fireworks×GPQA-Diamond is `BLOCKED_PROTOCOL_NONCONVERGENCE`: after parser/schema and token-floor repairs, the Fireworks-hosted model still exhausted its bounded generation budget on a material subset without producing a scorable MCQ answer. We record this as a protocol-level outcome: it is not treated as zero accuracy, not imputed, and not silently omitted (Section 6).

5.3. Adapter fidelity and provenance

L1/S1/TALE comparisons use closed-API adapters, not official training, model, or serving reproductions (Table 1; Appendix C). Each adapter preserves one budget-control surface that can be exercised through hosted APIs; Class A numerical attributions are therefore to the named adapters, not to the original systems. Azure×GPQA FTA is offline selector replay on stored generations. Trust tiers (Appendix D): Tier A denotes Cohere/Azure/Vertex×GSM8K cells with the strongest draw and scoring provenance; Tier B denotes the remaining completed cells. An earlier fallback-prompt SC attempt is excluded from all results; the Azure true same-model SC control records question-source hashes (Section 6.3).

Table 1: Baseline fidelity disclosure for closed-API adapters. Numerical claims in this manuscript apply to the evaluated adapter surfaces, not to the original L1, s1, or TALE systems.

Evaluated label	Source	Evaluated surface	Missing original component
L1-style adapter	Aggarwal & Welleck (2025)	Length-instruction prompt surface	No LCPO training/model
s1-inspired adapter	Muennighoff et al. (2025)	Chat "Wait" continuations	No local token-stream forcing
TALE-EP-style adapter	Han et al. (2025)	Prompt-budget injection with character heuristic	No zero-shot budget estimator; no TALE-PT

These rows are useful measurement objects because they expose budget-control surfaces under the same closed-API scoring protocol; they are not official reproductions of the source systems.

Repair asymmetry: bounded repair is overridden by Frontier's controller-committed answer but typically surfaces for L1/S1/TALE. Frontier's scored answer differs from its repair field on 151/3394 rows; external winner flips under the repair counterfactual: 0 (Appendix B).

5.4. Reproducibility levels

We distinguish three reproducibility claims. *Public aggregate-record verification:* released aggregate audit records, public manifests, source tables, and analysis scripts support aggregate-record verification of the reported tables and numerical invariants. *Historical API-output regeneration:* proprietary, stochastic closed-API services need not reproduce identical completions under the same public model ID. *Historical subset reconstruction:* for some Tier B cells the evaluated subset is preserved in aggregate form, but the complete original draw procedure cannot be reconstructed from the released artifact record; that incompleteness limits independent regeneration of the sampling step, not verification of the reported aggregate analyses (Appendix I). AI-assisted code inspection, code modification, and analysis organization were human-reviewed; all reported outputs were independently verified using frozen scripts, records, and manifests.

6. Matrix Landscape: Discovery, Adapters, and Scope

Table 2 is a workload landscape, not an efficiency or selector-attribution result. Under nominal budget B=6, it records Class A generators and Class B identical-pool selectors across providers and datasets; Section 5 defines the unscored Fireworks×GPQA protocol-blocked outcome. Sections 7–9 then isolate the comparisons the matrix alone cannot settle: identical-pool selection, provider transfer, and realized-resource accounting. Bounded repair

Table 2: Provider×dataset landscape under nominal B=6. Class A columns are closed-API generators (Frontier diagnostic; L1-/S1-/TALE-style adapters). Class B columns select over the *same* four-generator pool (pool cost $\leq 4B$; 0 extra model calls). Bold marks the highest Class A count and, separately, the higher of FTA vs. Pooled-4 as descriptive highlighting. Trust tiers and the Fireworks×GPQA blocked outcome are defined in Section 5.

Provider	Data	Front.	L1	S1	TALE	FTA	Pooled-4
Cohere	GSM8K	230/300	**249/300**	246/300	235/300	260/300	**261/300**
Cohere	MATH-500	**87/300**	73/300	84/300	76/300	92/300	**98/300**
Cohere	GPQA-D	68/198	74/198	69/198	**77/198**	**72/198**	**72/198**
Cohere	StrategyQA	78/100	69/100	77/100	**83/100**	**76/100**	**76/100**
Azure	GSM8K	**276/300**	269/300	246/300	204/300	260/300	**276/300**
Azure	MATH-500	**167/300**	142/300	161/300	112/300	143/300	**164/300**
Azure	GPQA-D	103/198	**105/198**	104/198	94/198	102/198	**104/198**
Azure	StrategyQA	70/100	72/100	**76/100**	74/100	**72/100**	**72/100**
Vertex	GSM8K	**279/300**	262/300	244/300	255/300	272/300	**278/300**
Vertex	MATH-500	217/300	220/300	210/300	**223/300**	**228/300**	224/300
Vertex	GPQA-D	**113/198**	106/198	108/198	110/198	**125/198**	123/198
Vertex	StrategyQA	**84/100**	82/100	83/100	**84/100**	**84/100**	**84/100**
Fireworks	GSM8K	**223/300**	210/300	160/300	191/300	224/300	**230/300**
Fireworks	MATH-500	100/300	105/300	94/300	**114/300**	**117/300**	**117/300**
Fireworks	GPQA-D	BLOCKED_PROTOCOL_NONCONVERGENCE					
Fireworks	StrategyQA	**81/100**	80/100	80/100	80/100	**79/100**	**79/100**

follows the historical convention in Section 4.4; Appendix B verifies that a uniform-repair counterfactual causes no external winner flips.

6.1. Does higher discovery predict safer selection?

Absent-from-pool errors are unrecoverable by pool-only selectors; present-but-misselected errors are the only errors a commit rule can still fix (Equation (3)). Writing DR for discovery rate and CSA for conditional selection accuracy (Equation (2); Figure 1), higher discovery does not systematically track FTA gains: global Pearson $r = -0.185$ (not significant at n=15; unstable under leave-one-out; supplementary scatter diagnostic). Azure×GSM8K has near-ceiling discovery with a large negative FTA delta, while Vertex and Azure GPQA share essentially the same discovery rate with opposite selector signs (Section 8). Joint DR/CSA reporting is therefore required before identical-pool tests.

6.2. What do Class A adapter counts characterize?

Descriptively, within the evaluated closed-API adapter surfaces, the reserved-attempt full-expansion diagnostic is sole highest in 7/15 cells and tied highest in one more; remaining adapter surfaces occupy fewer cells. We use these counts to characterize the workload surface before the controlled selection and accounting analyses. An offline incumbent-only counterfactual loses to full Frontier in 9/15 cells (Section 10), indicating that challenger expansion is not inert under stored traces.

6.3. What does the single-cell same-model control establish?

Table 3 is a single-cell supporting control on Azure GPT-4.1-mini with GSM8K (n=300): it matches call/sample budget between Frontier (B=6) and true same-model fixed-N self-consistency (SC; six samples). Each obtains 276/300 (92.00%; marginal exact binomial 95% CI 88.3304–94.8071%). Paired records show 298 ties and one method-only correct case each (two-sided exact McNemar p=1.0; paired-bootstrap 95% percentile interval for Frontier−SC: $[-1.00, +1.00]$ pp). No equivalence margin was prespecified. Panel B is contextual only.

7. Identical-Pool Selection

7.1. Does trace-conditioned selection beat identical-pool consensus?

Table 2's Class B columns and Figure 2 test whether failure-trace gated deferral (FTA) adds value beyond plurality when candidate availability is held fixed. FTA and Pooled-4 operate on the same four-generator answers, so their paired difference isolates the commit rule rather than discovery. On that shared pool (n=3394; 15 cells), Pooled-4 accuracy is 66.53% (2258/3394) and FTA is 65.00% (2206/3394): -1.53 pp for FTA (McNemar p=0.00027; discordant FTA/Pooled-4/tie counts 73/125/3196). A stratified within-cell paired bootstrap (10,000 replicates) places the Pooled-4−FTA gap at +1.53 pp with 95% percentile interval [+0.74, +2.33] pp; the gap remains positive under leave-one-provider, leave-one-dataset, and leave-one-cell deletion. Omitting Azure reduces the gap to +0.52 pp (McNemar p=0.30), so the aggregate advantage is concentrated but directionally stable. Descriptively, Pooled-4 matches or exceeds FTA in 13/15 completed cells (most per-cell gaps are within resampling noise; bootstrap replicates reproduce $\geq$ 13/15 only $\approx$ 17% of the time). FTA is strictly higher only

Table 3: Single-cell supporting control on Azure `gpt-4.1-mini`×GSM8K (n=300). Panel A matches call/sample budget (Frontier B=6 vs. SC with six samples); rows are not token-, latency-, dollar-, or FLOP-matched. Panel B is contextual adapter/FTA context on the same IDs.

Method	Operational class	Calls?	Pool/call basis	Correct	Total	Accuracy
Panel A: direct paired same-model control (common 300 examples)						
Frontier	Structured generation	Yes	Nominal budget B=6	276	300	92.00%
True same-model SC N=6	Fixed-N generation + vote	Yes	Exactly 6 samples	276	300	92.00%
Panel B: contextual adapter and replay rows (not a cost-matched class)						
L1-style adapter	Generation adapter	Yes	Nominal budget B=6	269	300	89.67%
FTA replay selector	Offline pool selector	No	Shared four-generator pool ($\leq 4B$); no selector calls	260	300	86.67%
S1-inspired adapter	Generation adapter	Yes	Nominal budget B=6	246	300	82.00%
TALE-EP-style adapter	Generation adapter	Yes	Nominal budget B=6	204	300	68.00%

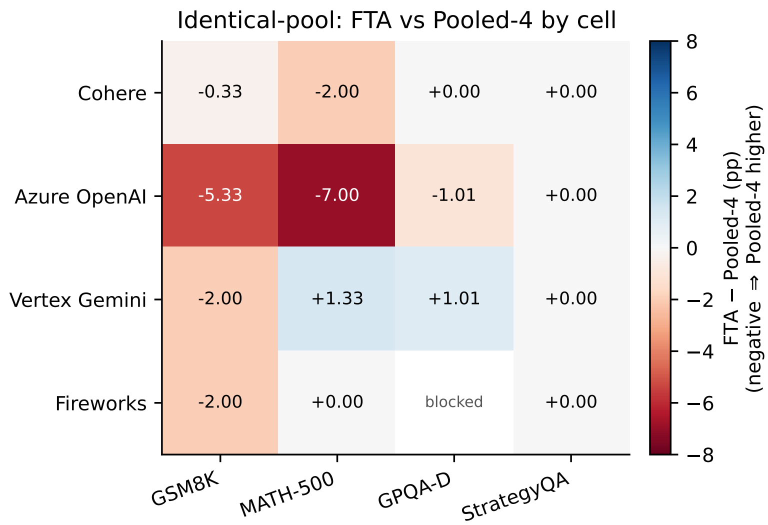

Figure 2: Per-cell difference in accuracy (FTA−Pooled-4, percentage points) on the *identical* four-generator pool. Negative values favor plurality. The comparison holds candidate availability fixed, so the map isolates commit-rule effects. Fireworks×GPQA is protocol-blocked.

on Vertex×MATH-500 and Vertex×GPQA. A unique plurality exists on 88% of examples; the 12% Frontier-fallback subset is harder (26% vs. 74% accuracy). Alternative deterministic tie rules move overall accuracy by at most ∼1.3 points (Appendix K).

Positive consensus regret CR(FTA) (Section 3) means this failure-derived rule extracts less accuracy from the shared pool than plurality. Without that control, occasional FTA gains over a single Class A generator could be misread as selection progress. The measurement point is attribution: once candidate availability is fixed, this commit rule has positive consensus regret despite some generator-relative gains.

7.2. When FTA differs from its Class A generator, is the pattern transfer?

Relative to Frontier alone, FTA improves in 8/15 cells, regresses in 6/15, and ties in 1/15. After Holm-corrected paired McNemar tests, supported gains appear in four cells and supported regressions in two (Azure×GSM8K, Azure×MATH-500). Figure 3 treats that heterogeneity as transfer evidence for the commit rule.

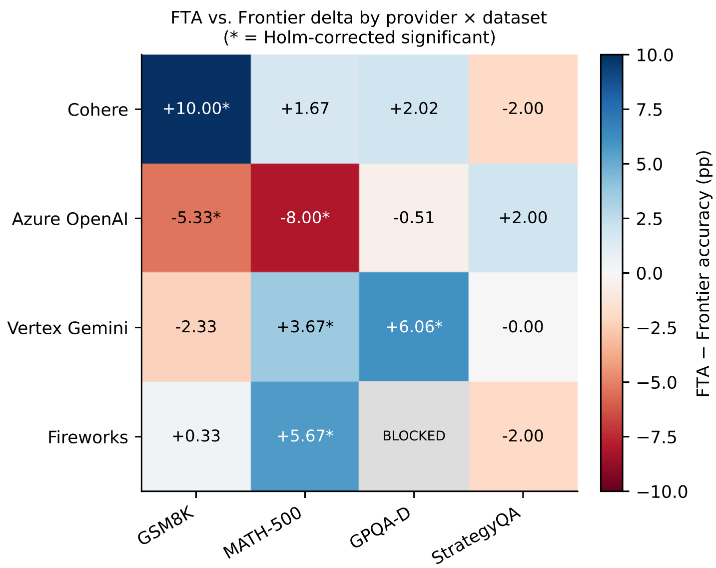

Figure 3: Per-cell FTA−Frontier accuracy delta. Asterisks mark Holm-corrected supported differences. Signs vary by provider; Azure has the only negative provider mean—transfer evidence for the commit rule, not a uniform selector ranking.

Table 4: Leave-one-cell-out selector over 15 completed cells with coarse cell-level features. The selector chooses Pooled-4 in every fold and ties fixed Pooled-4; the oracle row is nondeployable headroom.

Policy	Correct	Total	Accuracy	Deploy.	Outcome use
Held-out learned policy	2258	3394	66.53%	Yes	No
Best training-selected fixed method	2258	3394	66.53%	Yes	No
Fixed Pooled-4	2258	3394	66.53%	Yes	No
Frontier	2176	3394	64.11%	Yes	No
FTA replay selector	2206	3394	65.00%	Yes	No
Oracle cell selector	2287	3394	67.38%	No	Yes

Diagnostic cells. **Cohere×GSM8K** (Tier A; development-adjacent): high discovery (0.907); FTA improves over Frontier yet trails Pooled-4 slightly. **Azure×GSM8K**: discovery near ceiling (0.927) with FTA harming both Frontier and Pooled-4—selection failure under high discovery. **Vertex versus Azure GPQA:** matched discovery (≈ 0.636) with opposite FTA signs. StrategyQA (n=100) effects are near zero and are not treated as strong evidence.

7.3. Can cell-level selection improve over a fixed policy without leakage?

We ask whether a leakage-free selector calibrated only on training cells can choose among a frozen method set for a held-out provider–dataset cell. The design uses leave-one-cell-out over the 15 completed cells; candidate methods are Frontier, FTA, the L1-/S1-/TALE-style adapters, and Pooled-4; features are coarse cell-level summaries (provider, dataset, and training-only empirical accuracies). The held-out protocol was fixed before the reported evaluation and archived with the supplementary diagnostics. An earlier non-leakage-free analysis is excluded.

Table 4 gives the full result. Within this 15-cell design and feature set, the leakage-free selector does not improve over fixed Pooled-4. This negative result is informative for attribution: the available coarse cell-level features did not identify a deployable selector more reliable than a fixed identical-pool plurality rule, while the oracle row shows the nondeployable headroom that remains when held-out outcomes are known.

Table 5: Per-cell rescue/regression counts for FTA's gates (FIX-2 / FIX-4). Applied switches: FIX-2 606, FIX-4 21, overlap 0. The table characterizes transfer of this failure-mined commit rule.

Provider	Data	F2 trig.	F2 resc.	F2 regr.	F4 trig.	F4 resc.	F4 regr.
Cohere	GSM8K	63	36	9	3	3	0
Cohere	MATH-500	119	20	15	1	0	0
Cohere	GPQA-D	43	11	7	4	2	2
Cohere	StrategyQA	0	0	0	2	0	2
Azure	GSM8K	26	2	18	0	0	0
Azure	MATH-500	60	7	31	2	0	0
Azure	GPQA-D	33	8	8	3	1	2
Azure	StrategyQA	0	0	0	4	3	1
Vertex	GSM8K	17	4	11	0	0	0
Vertex	MATH-500	20	11	0	0	0	0
Vertex	GPQA-D	37	16	4	0	0	0
Vertex	StrategyQA	0	0	0	0	0	0
Fireworks	GSM8K	61	19	18	0	0	0
Fireworks	MATH-500	125	26	9	0	0	0
Fireworks	StrategyQA	2	1	1	2	0	2

8. Provider Heterogeneity and Transfer

Section 7 showed that FTA trails identical-pool plurality in aggregate. Here we ask whether that gap is explained by unreliable transfer of FTA's failure-mined gates across providers. Table 5 reports per-cell rescue/regression counts for the two constituent gates (Low-Depth Guard / FIX-2; External-Consensus Fallback / FIX-4; Section 4.3).

8.1. Provider-stratified sign patterns

Mean FTA−Frontier deltas by provider: Cohere positive, Azure negative, Vertex and Fireworks mildly positive (Fireworks has three completed cells). Azure is the only provider with a negative mean; within Azure, harm concentrates on GSM8K/MATH-500 rather than StrategyQA or GPQA.

8.2. Which gate drives the movement?

Offline gate-only replay confirms FIX-2 accounts for essentially all FTA movement (606 applied switches vs. 21 for FIX-4; zero overlap; FIX-2-only accuracy matching FTA globally; Appendix G). Cell-level counts diverge: Azure×GSM8K FIX-2 yields 2 rescues against 18 regressions, and Azure×MATH-500 FIX-2 yields 7 rescues against 31 regressions,

Table 6: Mean realized resources per evaluated Class A policy across 15 cells (nominal B=6). Class B pool cost inherits up to four Class A pipelines (Table 2). USD uses a 2026-07-16 pricing snapshot. Frontier token/cost reconstruction status is documented in the table note.

Class A method	Tokens/ex.†	Latency (s)	Calls (nom.=6)	Cost/ex.† ($)
Frontier	1287.0†	16.87	5.38	0.00258†
L1-style adapter	728.7	5.48	1.61	0.00133
S1-inspired adapter	1236.0	8.80	2.85	0.00234
TALE-EP-style adapter	727.3	5.33	1.66	0.00129

†Frontier tokens/cost are a documented lower bound (challenger-expansion-phase tokens are not recoverable from historical records without fabrication); calls, latency, and all adapter figures are complete. See Section 9.

whereas the same Low-Depth Guard is net-positive on development-adjacent Cohere×GSM8K and on Fireworks×MATH-500 (Table 5). Aggregating the same frozen cell summaries localizes the transfer failure: FIX-2 is modestly net-positive overall (161 rescues, 131 regressions over 606 switches) but nearly neutral after removing the development-adjacent Cohere×GSM8K cell (125 rescues, 122 regressions over 543 switches). The two Azure mathematics cells account for most of the harm (9 rescues, 49 regressions over 86 switches; rescue/regression rates 10.5%/57.0%).

Thus FTA is mechanistically FIX-2-dominated, and Azure mathematics cells are concrete transfer failures for this failure-mined commit rule under the evaluated provider shifts. FIX-2/FIX-4 were developed from Cohere×GSM8K failure diagnostics and frozen before the remaining cells, so the Azure and other non-development cells are the stronger transfer evidence (Section 11).

9. Compute Accounting and Instrumentation Semantics

Table 6 and Figure 4 evaluate the accounting semantics of equal Class A nominal budget B=6. Following Section 3, we separate (i) nominal logical actions, (ii) successful provider completions on the reported path, (iii) internal HTTP retries, (iv) generated-token fields, (v) observed latency, and (vi) estimated dollar cost. The columns are intentionally not collapsed: an instrumentation defect can change resource conclusions while leaving extracted answers and accuracy unchanged.

Table 6 reports unweighted means of per-cell means, and Table 7 records

Table 7: Accounting status for resource fields used in Table 6 and Figure 4. The status labels distinguish exact, reconstructed, incomplete, and lower-bound quantities; no missing token, retry, or dollar fields are imputed.

Field	Accounting status	Interpretation
Nominal budget	Fixed at B=6 for each Class A policy; not a realized resource.	Equalizes logical opportunity only.
Successful calls	Adapter calls observed; Frontier calls reconstructed from stored outer calls plus maturity.	Exact after reconstruction; mean Frontier calls change from 2.78 to 5.38.
Tokens and USD	Adapter token fields/pricing are complete; Frontier challenger-phase tokens are absent.	Frontier token and dollar ratios are lower bounds; Figure 4 is not a Pareto frontier.
Latency	Reported as a separate Class A realized-resource field.	Not collapsed with accuracy, tokens, or dollars.
Retries	Provider/harness retry records are partial; paid-retry accounting is incomplete.	Retry totals are not used as complete cost claims.

which resource fields support exact versus lower-bound interpretations. Class B end-to-end cost inherits up to four Class A pipelines.

Instrumentation audit and successful-call reconstruction. A harness gap reset/read usage on Frontier's outer generator only, omitting the challenger-expansion generator. On preserved validated Frontier records, successful calls reconstruct as outer successful calls plus `frontier_candidate_maturity` (maturity = challenger expansions + verifier calls). Mean successful calls rise from 2.78 to 5.38 of nominal B=6 (example-weighted 5.44). Ratios versus the evaluated adapters: Frontier/L1-style 2.13×–5.18×, Frontier/TALE-style 2.03×–5.51×, Frontier/S1-inspired 1.43×–2.50× (Frontier higher in every cell). Challenger-phase tokens were never persisted, so Frontier token and USD ratios are lower bounds (1.22×–2.56× vs. L1-style; 1.13×–2.93× vs. TALE-style; mixed vs. S1-inspired); paid retries are likewise incomplete. Accuracy and McNemar/Holm tests are unchanged. The audit supports the claim that matched B=6 did not produce matched realized call allocations (Appendix K).

Figure 4 is therefore a diagnostic for incomplete cost telemetry, not a definitive cost-efficiency frontier. The single-cell Azure×GSM8K SC comparison (Section 6.3) remains call/sample-budget-matched only.

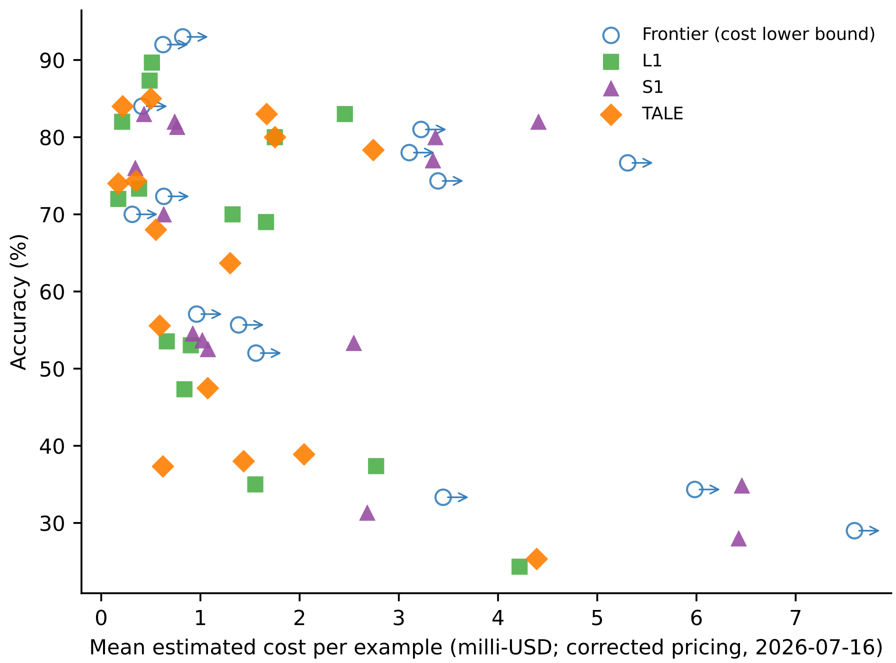

Figure 4: Incomplete-telemetry diagnostic for Class A accuracy and mean estimated cost per example (milli-USD). Open Frontier markers with rightward arrows indicate that challenger-phase token fields were not persisted, so Frontier cost positions are lower bounds. This diagnostic should not be read as a Pareto frontier: policies are not resource-matched, and paid retries are incomplete.

10. Offline Counterfactuals

Frozen-trace counterfactuals isolate attribution questions that aggregate accuracy cannot answer: whether challenger expansion changes outcomes relative to the incumbent, and whether repair scoring—orthogonal to allocation—moves commits or external rankings (Section 6). Gate-level transfer evidence remains in Section 8. Historical variants and diagnostics outside the current protocol are documented only in the supplementary material.

10.1. Incumbent versus full Frontier

Using stored incumbent answers from controller metadata, full Frontier beats the incumbent-only counterfactual in 9/15 cells and loses in 2/15 (Appendix H). This is a discovery-side non-inertness check under stored traces, not a matched-cost "no expansion" ablation.

10.2. Repair scoring counterfactual

Bounded repair is not credited to Frontier allocation. On the completed 15-cell matrix, Frontier's scored answer differs from its repair-related field on 151/3394 rows; forcing the repair field for Frontier would yield 0 rescues and 68 regressions in the available counterfactual, while a repair-based external ranking counterfactual flips 0 external winners (Appendix B). These outcomes separate scoring convention from selection and accounting; primary tables retain the historical scoring convention.

11. Limitations

Environments and budget. Evidence covers four closed-API providers and four reasoning datasets at one nominal budget B=6, with 15/16 cells completed and Fireworks×GPQA protocol-blocked (Section 5). Aggregate identical-pool tests use n=3394 paired examples, but transfer and cell-level policy claims rest on fewer independent environments; StrategyQA is underpowered for small deltas. Algorithm rankings at other budgets are not claimed. Natural Plan appears only as a supplementary pilot record.

Baseline fidelity and development adjacency. L1/S1/TALE rows are closed-API adapters, not official reproductions (Table 1). The main provenance qualifications are stated in Section 5; historical replay notes are supplementary. Metrics assume extractable, normalizable final answers.

Resource telemetry. Section 9 gives the resource-accounting detail. Token/USD fields remain lower bounds, paid retries are incomplete, and the study does not report FLOP matching or a full resource-matched matrix. Public-benchmark contamination in proprietary training data cannot be audited.

Reproducibility and temporal validity. Released artifacts support aggregate-record verification of reported tables and numerical invariants (Section 5.4). They do not guarantee regeneration of identical proprietary outputs. Closed-API model identifiers do not necessarily provide immutable weight-level versioning, so provider-side updates may limit exact longitudinal regeneration. Historical component diagnostics are not ablations of the evaluated protocol.

12. Discussion

The protocol is intended for evaluations where the performance object is broader than final accuracy: multi-provider API studies, candidate-pool and selector studies, router or cascade studies, closed commercial APIs, incomplete telemetry, failed or unscorable cells, and cost- or latency-sensitive inference. In those settings, the central practical lesson is to report the layer at which each claim is measured. A nominal call budget, the realized resource vector, the cost of constructing a candidate pool, and the marginal cost of a selector are different performance objects.

Compare selectors on fixed candidate pools. When a study evaluates a nontrivial commit rule, report accuracy against plurality on the same extracted answers and disclose the pool-construction cost. This separates selection from discovery and prevents generator-relative gains from being mistaken for selector progress (Sections 7 and 7.3).

Treat failure-mined rules as transfer claims. Rules developed from one provider or workload should carry a development-environment label and be reported with provider-stratified rescue/regression counts. The relevant question is not only whether the rule helps somewhere, but whether its failure signature transfers under the evaluated provider shifts (Section 8).

Report nominal budget with componentwise realized resources. Equal nominal budgets do not by themselves equalize successful completions, retries, generated tokens, latency, or dollars. Resource fields should therefore be reported componentwise, with reconstruction status and lower-bound fields marked explicitly (Section 9).

Preserve blocked outcomes and discovery–selection structure. Protocol-blocked cells should remain visible rather than being scored as zero or silently removed. For completed cells, joint reporting of discovery rate and conditional selection accuracy identifies whether errors are absent-from-pool failures or present-but-misselected failures (Sections 6 and 5).

These practices are actionable for future closed-API evaluations because they do not require access to provider internals. They require disciplined accounting: the evaluation must say which resources were observed, which were reconstructed, which remain lower bounds, which cells failed the bounded protocol, and which selector comparisons share the same candidate pool.

13. Conclusion

This article contributes a performance-evaluation protocol for closed-API, budgeted LLM inference. The protocol joins nominal-budget accounting, realized-resource measurement, discovery–selection separation, identical-pool selector controls, provider-transfer labels, provenance tiers, protocol-blocked outcomes, and incomplete-telemetry disclosure in one auditable evaluation contract.

The evidence shows why those separations matter: selector conclusions change when candidate pools are fixed, failure-mined gates can transfer poorly across providers, and reconstructed successful completions can change efficiency interpretation without changing accuracy. The paper does not claim a universal algorithm ranking, a complete cost-efficiency frontier, or identical regeneration of proprietary API outputs. Its claim is methodological: future multi-provider, selector, router, cascade, or cost-sensitive closed-API evaluations need to state the resource layer and attribution control that each performance conclusion depends on.

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Reproducibility Statement

Released aggregate audit records, public manifests, source tables, and analysis scripts support aggregate-record verification of the reported tables and numerical invariants for all 15 completed cells; provenance tiers and replay limits are documented in Appendix I. The released artifacts do not guarantee regeneration of identical proprietary API outputs, and live regeneration may incur API charges. The manifests define the public aggregate-record verification boundary for the reported analyses.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

GSM8K, MATH-500, GPQA-Diamond, and StrategyQA are publicly available from their original sources. The public repository contains the evaluation code, frozen aggregate audit records, analysis scripts, manifests, and reproducibility instructions: `https://github.com/SoroushVahidi/frontier-allocation-for-budgeted-llm-inference`. Underlying per-example proprietary API outputs and complete historical subset records are not publicly shareable or fully reconstructable; the released aggregate records therefore support verification of reported aggregates rather than live output regeneration. The fixed public artifact commit is:

`dfc9997d803199d699e23ee42cfe0777e6d78155`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, the author used ChatGPT, Claude, and Cursor to assist with drafting and revising prose. Research-process uses of AI-assisted tools for code inspection, code modification, and analysis organization are described in Section 5.4. The author reviewed and edited the tool-assisted prose as needed and takes full responsibility for the content of the published article.

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This appendix supplies disclosure and provenance detail summarized in the main text: gold-free features, repair and baseline fidelity, trust tiers, statistical testing, pseudocode, offline counterfactuals, and compute detail. Historical negative variants, pilot surfaces, and partial same-model checks are moved to the supplementary material because they are outside the evaluated protocol.

Appendix A. Gold-Free Runtime Features

Every quantity used by Frontier (incumbent support, challenger support, override margin, and maturity) and by FTA (trace trigger, external agreement) is computed from generated candidates only. Gold labels are offline-only for accuracy and discovery/selection diagnostics. Runtime leakage checks found no gold-dependent runtime features. Researcher-adaptation concerns remain separate from runtime leakage (Section 11).

Appendix B. Bounded Answer Repair

Bounded repair re-ranks completed branch nodes by canonicalized answer support. Surfacing overrides repair with a controller-committed answer when present. Frontier almost always supplies a committed answer, so repair is computed but discarded for scoring; L1/S1/TALE typically score the repair-layer output. Offline quantification on the 15-cell matrix: Frontier's scored answer differs from the repair-related field on **151** of 3,394 Frontier rows. Among Frontier differs, hypothetical repair application yields 0 rescues and 68 regressions in this reconstruction (remainder wrong↔wrong / neutral). External `repair_answer_raw` vs. scored often differs textually (462 rows) largely due to canonicalization; forcing a repair-based external ranking counterfactual produced **0** external winner flips. Asymmetry remains real; external cell winners appear ranking-stable under this counterfactual check.

Appendix C. Baseline Fidelity (Full)

L1-style adapter. Original contribution is LCPO training; we use only the length-instruction prompt surface (closer to the paper's untrained budget-prompt ablation).

S1-inspired adapter. Original suppresses end-of-thinking in a local token stream; we approximate via repeated chat "Wait" continuations.

TALE-EP-style adapter. We target TALE-EP prompt injection; replace zero-shot budget estimation with a deterministic character-length heuristic to preserve the B=6 call-budget protocol. TALE-PT is not reproduced.

Appendix D. Provenance and Trust Tiers

Tier A: Cohere/Azure/Vertex×GSM8K (strongest draw and scoring provenance under the released manifests). Tier B: remaining completed cells. Three cells had historical duplicate-row noise; deduplication changed 0 accuracies. Azure×GPQA FTA: offline selector replay on stored generations. Fireworks×GPQA: protocol-blocked (`BLOCKED_PROTOCOL_NONCONVERGENCE`). For some Tier B cells the frozen evaluated subset is preserved even when the complete historical draw procedure cannot be reconstructed (Section 5.4).

Appendix E. Statistical Testing

Paired exact McNemar tests; Holm–Bonferroni within relevant comparison families. Class A and Class B are not Holm-mixed as if they shared a budget unit. Bootstrap vs. McNemar disagreement on Cohere×GSM8K FTA-vs-L1 (high ties) is reported historically; we do not revive the weaker significance claim. For Pooled-4 vs. FTA, the stratified within-cell paired bootstrap on the frozen n=3394 rows gives a 95% percentile interval [+0.74, +2.33] pp. Provider- and dataset-level leave-one-out deletions are sensitivity checks on the same sample, not independent replications.

Appendix F. Procedures (Pseudocode)

Appendix F.1. Frontier (Class A reserved-attempt full-expansion policy)

Table F.8 lists the fixed public parameters for the reported matrix cells.

Frontier procedure.

1. **Input:** instance x and nominal budget B=6.
2. Generate two reserved direct-chain attempts with distinct prompt styles.
3. Group extracted answers by the public gold-free canonicalization rule and choose the incumbent by plurality; ties break by first-observed answer group.

Table F.8: Frontier public parameters for all reported matrix cells. These are fixed across providers and datasets.

Quantity	Executed value
Nominal budget	B=6
Reserved direct attempts	2 prompt styles
Answer grouping	Gold-free canonical answer key; first-observed tie-break
Template early gate	support ≥ 2.0, margin ≥ 2.0, entropy ≤ -1.0 (sentinel)
Gate status in reported runs	Disabled by sentinel; never fired
Override maturity threshold	challenger maturity ≥ 2
Override support-margin threshold	challenger support $-$ incumbent support ≥ 1
Override guard	no override when incumbent appears in challenger-supported groups

4. Allocate the remaining nominal budget, if any, to challenger generation and verification. The template retained an early-commit gate, but the evaluated thresholds are sentinel values (support threshold 2.0 with vote-share support in $[0, 1]$), so early commitment never fires and is not part of the effective evaluated policy.
5. Override the incumbent with the challenger iff a challenger exists, differs from the incumbent, has support greater than one, has maturity ≥ 2, has support margin ≥ 1, and the incumbent is not already represented among challenger-supported groups.
6. **Output:** the committed answer. All thresholds and tie rules are fixed across cells; features are gold-free.

Action semantics. For Frontier, one nominal action is one direct-chain attempt or one challenger expansion step, each corresponding to one successful provider response on the reported path. For L1 and TALE, one action is one length/budget-conditioned generation attempt; for S1, it is the initial generation or a forced "Wait" continuation. Canonicalization, plurality, repair ranking, FTA gates, and Pooled-4 voting issue no model calls. Internal HTTP retries are counted in retry telemetry but not as new logical actions unless a new successful response is produced.

Pooled-4 (Class B consensus).

1. **Input:** committed answers of Frontier, L1, S1, and TALE for instance x.

2. Let $a^\star$ be a plurality mode (agreement ≥ 2).
3. If no unique mode, return Frontier's answer (deterministic tie-break).
4. **Output:** $a^\star$. No additional model calls.

FTA (Class B gated deferral).

1. **Input:** Frontier answer and trace metadata; L1/S1/TALE answers.
2. If Low-Depth Guard (FIX-2) triggers on a single-weak-branch (or support-zero) signature, return majority among $\{\mathrm{L1}, \mathrm{S1}, \mathrm{TALE}\}$ (agreement ≥ 2; priority TALE$\succ$S1$\succ$L1).
3. Else if External-Consensus Fallback (FIX-4) triggers (direct incumbent agreement and unanimous external disagreement with Frontier), return the unanimous external answer.
4. Else return Frontier's answer.
5. Gates are mutually exclusive by construction; at most one fires.

Appendix G. Gate Details and Ablation

FIX-2 and FIX-4 are non-overlapping by construction. Applied switches: 606 / 21; overlap 0. FIX-2-only accuracy equals FTA globally; FIX-4-only $\approx$ Frontier. Azure harm is FIX-2-driven (Section 8).

Appendix H. Incumbent Ablation

Using stored incumbent-only answers from controller metadata: Frontier beats incumbent in 9/15 cells; incumbent better in 2/15. Large gains on several MATH cells. Caveat: metadata completeness varies. This is an offline comparison against stopping after the reserved attempts, not a live early-stopping policy and not a matched-cost alternative.

Appendix I. Reproducibility Records

A public claim–evidence map links numerical claims to tracked aggregate audit records, source tables, analysis scripts, and package checksums for the Azure true-SC comparison, held-out selector, and 15-cell matrix summaries. The public artifacts support aggregate-record verification of the reported numerical invariants, not identical proprietary regeneration or every historical subset draw (Section 5.4). Live reruns would require provider credentials and paid API calls and are not part of this manuscript.

Appendix J. Supplementary Diagnostics

The descriptive discovery–selection scatter diagnostic is provided in the supplementary material; the tested transfer evidence used in the main paper is reported in Section 8.

Appendix K. Additional Compute Detail

Frontier/L1: tokens 1.10×–2.57× (lower bound), calls 2.13×–5.18× (exact), latency 2.11×–4.39×. Frontier/TALE: tokens 1.01×–2.65× (lower bound), calls 2.03×–5.51× (exact), latency 1.80×–6.77×. Frontier/S1: tokens mixed (lower bound); calls 1.43×–2.50× (exact; Frontier higher every cell); latency higher every cell. Dollar figures use recomputed pricing from measured tokens (2026-07-16 snapshot). Tokens, USD from incomplete token fields, and paid retries remain lower bounds.

Pooled-4 tie-rule sensitivity. On 3,259 complete four-method quads, 88.0% have a unique plurality and 12.0% fall back. Accuracy under Frontier-fallback (67.8% on this reconstruction) moves to 66.7%/66.5% under two predeclared alternatives (≤ 1.3 pp); none selected post hoc. This reconstruction excludes 135 incomplete-quad rows and is for tie-sensitivity only, not a replacement of the 66.53%/n=3394 aggregate.